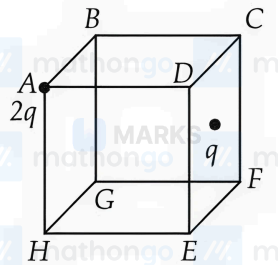
**Q1. JEE Main 2026 (21 January Shift 2)**

Consider two identical metallic spheres of radius $R$ each having charge $Q$ and mass $m$. Their centers have an initial separation of $4R$. Both the spheres are given an initial speed of $u$ towards each other. The minimum value of $u$, so that they can just touch each other is :
(Take $k = \frac{1}{4\pi\epsilon_0}$ and assume $kQ^2 > Gm^2$ where $G$ is the Gravitational constant)

(1) $\sqrt{\frac{kQ^2}{2mR}\left(1 - \frac{Gm^2}{kQ^2}\right)}$
(2) $\sqrt{\frac{kQ^2}{4mR}\left(1 - \frac{Gm^2}{kQ^2}\right)}$
(3) $\sqrt{\frac{kQ^2}{2mR}\left(1 - \frac{Gm^2}{2kQ^2}\right)}$
(4) $\sqrt{\frac{kQ^2}{4mR}\left(1 + \frac{Gm^2}{kQ^2}\right)}$

**Q2. JEE Main 2026 (23 January Shift 1)**

Two point charges $2q$ and $q$ are placed at vertex $A$ and centre of face $CDEF$ of the cube as shown in figure. The electric flux passing through the cube is :




(1) $\frac{3q}{2\epsilon_0}$
(2) $\frac{3q}{4\epsilon_0}$
(3) $\frac{q}{\epsilon_0}$
(4) $\frac{3q}{\epsilon_0}$

**Q3. JEE Main 2026 (28 January Shift 2)**

Identify the correct statements :

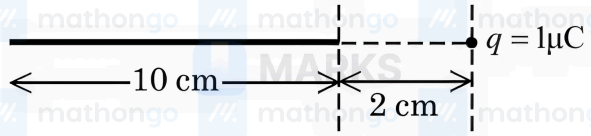
A. Electrostatic field lines form closed loops.
B. The electric field lines point radially outward when charge is greater than zero.
C. The Gauss - Law is valid only for inverse - square force.
D. The workdone in moving a charged particle in a static electric field around a closed path is zero.
E. The motion of a particle under Coulomb's force must take place in a plane.

Choose the correct answer from the options given below :

(1) A, B, D, E Only
(2) A, C, E Only
(3) A, B, C, D Only
(4) B, C, D, E Only

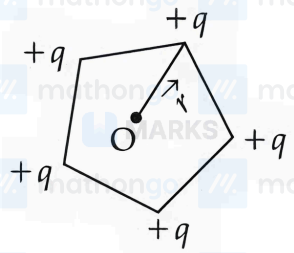
**Q4. JEE Main 2026 (24 January Shift 2)**

A point charge $q = 1\mu C$ is located at a distance 2 cm from one end of a thin insulating wire of length 10 cm having a charge $Q = 24\mu C$, distributed uniformly along its length, as shown in figure. Force between $q$ and wire is ____ N. (Use : $\frac{1}{4\pi\epsilon_o} = 9 \times 10^9$ N. m$^2$/C$^2$)




**Q5. JEE Main 2026 (22 January Shift 2)**

Five positive charges each having charge $q$ are placed at the vertices of a pentagon as shown in the figure. The electric potential $(V)$ and the electric field $(\vec{E})$ at the center $O$ of the pentagon due to these five positive charges are:




(1) $V = \frac{5q}{4\pi\varepsilon_o r}$ and $\vec{E} = \frac{5\sqrt{3}q}{8\pi\varepsilon_o r^2}\hat{r}$

(2) $V = \frac{5q}{4\pi\varepsilon_o r}$ and $\vec{E} = 0$

(3) $V = 0$ and $\vec{E} = 0$

(4) $V = \frac{5q}{4\pi\varepsilon_o r}$ and $\vec{E} = \frac{5q}{4\pi\varepsilon_o r^2}\hat{r}$

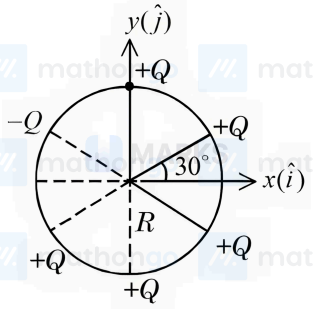
**Q6. JEE Main 2026 (22 January Shift 2)**

Three small identical bubbles of water having same charge on each coalesce to form a bigger bubble. Then the ratio of the potentials on one initial bubble and that on the resultant bigger bubble is :

(1) $3^{2/3} : 1$ (2) $1 : 2^{2/3}$ (3) $1 : 3^{2/3}$ (4) $1 : 3^{1/3}$

**Q7. JEE Main 2026 (22 January Shift 1)**

Six point charges are kept 60° apart from each other on the circumference of a circle of radius $R$ as shown in figure. The net electric field at the center of the circle is ____. ($\epsilon_0$ is permittivity of free space)




(1) $\frac{Q}{4\pi\epsilon_0 R^2}(\sqrt{3}\hat{i}-\hat{j})$
(2) $-\frac{5Q}{8\pi\epsilon_0 R^2}(\hat{i}+\sqrt{3}\hat{j})$
(3) $-\left(\frac{5Q}{8\pi\epsilon_0 R^2}\right)(\hat{i}-3\hat{j})$
(4) $-\frac{Q}{4\pi\epsilon_0 R^2}(\sqrt{3}\hat{i}-\hat{j})$

**Q8. JEE Main 2026 (22 January Shift 1)**

A simple pendulum has a bob with mass $m$ and charge $q$. The pendulum string has negligible mass. When a uniform and horizontal electric field $\vec{E}$ is applied, the tension in the string changes. The final tension in the string, when pendulum attains an equilibrium position is ____. (g: acceleration due to gravity)

(1) $\sqrt{m^2g^2+q^2E^2}$
(2) $m\,g+qE$
(3) $mg-qE$
(4) $\sqrt{m^2g^2-q^2E^2}$

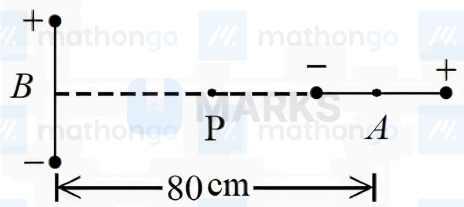
**Q9. JEE Main 2026 (22 January Shift 1)**

Electric field in a region is given by $\vec{E}=Ax\hat{i}+By\hat{j}$, where $A=10\ \mathrm{V/m^2}$ and $B=5\ \mathrm{V/m^2}$. If the electric potential at a point $(10,20)$ is 500 V, then the electric potential at origin is ____ V.

(1) 0
(2) 2000
(3) 500
(4) 1000

**Q10. JEE Main 2026 (23 January Shift 2)**

Two shorts dipoles $(A,B)$, $A$ having charges $\pm 2\mu$C and length 1 cm and $B$ having charges $\pm 4\mu$C and length 1 cm are placed with their centres 80 cm apart as shown in the figure. The electric field at a point $P$, equi-distant from the centres of both dipoles is ____ N/C.




(1) $\frac{9}{16}\sqrt{2}\times 10^4$
(2) $9\sqrt{2}\times 10^4$
(3) $4.5\sqrt{2}\times 10^4$
(4) $\frac{9}{16}\sqrt{2}\times 10^5$

**Q11. JEE Main 2026 (23 January Shift 2)**

Two charges $7\mu$C and $-2\mu$C are placed at $(-9,0,0)$cm and $(9,0,0)$cm respectively in an external field $E=\frac{A}{r^2}\hat{r}$, where $A=9\times10^5$ N/C. m$^2$. Considering the potential at infinity is 0, the electrostatic energy of the configuration is ____ J.

(1) 49.3 (2) -90.7 (3) 24.3 (4) 1.4

**Q12. JEE Main 2026 (21 January Shift 1)**

A point charge of $10^{-8}$C is placed at origin. The work done in moving a point charge $2\mu$C from point $A(4,4,2)$m to $B(2,2,1)$m is ____ J. $\left(\frac{1}{4\pi\epsilon_o}=9\times10^9 \text{ in SI units }\right)$

(1) $45\times10^{-6}$ (2) $15\times10^{-6}$
(3) 0 (4) $30\times10^{-6}$

**Q13. JEE Main 2026 (28 January Shift 1)**

Two point charges of 1 nC and 2 nC are placed at the two corners of equilateral triangle of side 3 cm. The work done in bringing a charge of 3 nC from infinity to the third corner of the triangle is ____ $\mu$J.
$\frac{1}{4\pi\epsilon_0}=9\times10^9$ N. m$^2$/C$^2$

(1) 27 (2) 5.4 (3) 2.7 (4) 3.3

**Q14. JEE Main 2026 (24 January Shift 1)**

There are three co-centric conducting spherical shells $A, B$ and $C$ of radii $a, b$ and $c$ respectively $(c>b>a)$ and they are charged with charge $q_1, q_2$ and $q_3$ respectively. The potentials of the spheres $A, B$ and $C$ respectively, are :

(1) $\frac{1}{4\pi\epsilon_o}\left(\frac{q_1+q_2+q_3}{a}\right), \frac{1}{4\pi\epsilon_o}\left(\frac{q_1+q_2+q_3}{b}\right), \frac{1}{4\pi\epsilon_o}\left(\frac{q_1+q_2+q_3}{c}\right)$
(2) $\frac{1}{4\pi\epsilon_o}\left(\frac{q_1}{a}+\frac{q_2}{b}+\frac{q_3}{c}\right), \frac{1}{4\pi\epsilon_o}\left(\frac{q_1+q_2+q_3}{b}\right), \frac{1}{4\pi\epsilon_o}\left(\frac{q_1+q_2+q_3}{c}\right)$
(3) $\frac{1}{4\pi\epsilon_o}\left(\frac{q_1}{a}+\frac{q_2}{b}+\frac{q_3}{c}\right), \frac{1}{4\pi\epsilon_o}\left(\frac{q_1+q_2}{b}+\frac{q_3}{c}\right), \frac{1}{4\pi\epsilon_o}\left(\frac{q_1+q_2+q_3}{c}\right)$
(4) $\frac{1}{4\pi\epsilon_o}\left(\frac{q_1+q_2+q_3}{a}\right), \frac{1}{4\pi\epsilon_o}\left(\frac{q_1+q_2}{b}+\frac{q_3}{c}\right), \frac{1}{4\pi\epsilon_o}\left(\frac{q_1}{a}+\frac{q_2}{b}+\frac{q_3}{c}\right)$

**Q15. JEE Main 2026 (24 January Shift 1)**

Three charges $+2q, +3q$ and $-4q$ are situated at $(0,-3a), (2a,0)$ and $(-2a,0)$ respectively in the $xy$ plane. The resultant dipole moment about origin is ____.

(1) $2qa(7\hat{i}-3\hat{j})$ (2) $2qa(3\hat{j}-\hat{i})$
(3) $2qa(3\hat{i}-7\hat{j})$ (4) $2qa(3\hat{j}-7\hat{i})$

**Q16. JEE Main 2026 (24 January Shift 1)**

The electrostatic potential in a charged spherical region of radius $r$ varies as $V=ar^3+b$, where $a$ and $b$ are constants. The total charge in the sphere of unit radius is $\alpha\times\pi a\in_o$. The value of $\alpha$ is ____.
(permittivity of vacuum is $\epsilon_0$)

(1) $-9$ (2) -12 (3) -8 (4) -6

## ANSWERS AND SOLUTIONS

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **1.** (2) | **2.** (2) | **3.** (4) | **4.** 90 | **5.** (2) | **6.** (3) | **7.** (4) | **8.** (1) |
| **9.** (2) | **10.** (1) | **11.** (1) | **12.** (4) | **13.** (3) | **14.** (3) | **15.** (1) | **16.** (2) |

**1.** **(2)** Using energy conservation, the system's kinetic energy must overcome the increase in electrostatic potential energy as the spheres approach.
Initial state: separation 4R, speed u each.
Final state: touching (separation 2R), speed zero.
Change in potential energy: $\Delta U = k\frac{Q^2}{2R} - k\frac{Q^2}{4R} = k\frac{Q^2}{4R}$.
Initial kinetic energy: $KE_i = mu^2$.
Energy equation: $mu^2 = k\frac{Q^2}{4R}\left(1 - \frac{Gm^2}{kQ^2}\right)$.
Therefore, $u = \sqrt{\frac{kQ^2}{4mR}\left(1 - \frac{Gm^2}{kQ^2}\right)}$.

**2.** **(2)** Using Gauss's law:
$\phi = \frac{Q_{in}}{\epsilon_0}$

Charge $2q$ is at vertex A of the cube. A vertex is shared by 8 cubes, so the flux contribution from $2q$ through this cube is $\frac{2q}{8\epsilon_0} = \frac{q}{4\epsilon_0}$.

Charge $q$ is at the centre of face CDEF. A face center is shared by 2 cubes, so the flux contribution from $q$ through this cube is $\frac{q}{2\epsilon_0}$.

Total flux:
$\phi = \frac{q/4+q/2}{\epsilon_0} = \frac{3q}{4\epsilon_0}$

**3.** **(4)** Statement A: Electrostatic field lines originate from positive charges and terminate on negative charges. They do not form closed loops because the electrostatic field is conservative. Thus, A is incorrect.
Statement B: For a positive point charge ($q > 0$), the electric field lines are directed radially outwards. Thus, B is correct.
Statement C: Gauss's Law is a consequence of the inverse-square nature of Coulomb's law ($F \propto 1/r^2$). If the force law were different, Gauss's Law would not hold in its standard form. Thus, C is correct.
Statement D: The electrostatic field is conservative, meaning the work done in moving a charge along any closed path is zero ($\oint \vec{E} \cdot d\vec{l} = 0$). Thus, D is correct.
Statement E: Coulomb's force is a central force. The motion of a particle under a central force always takes place in a plane because the torque about the center is zero, keeping angular momentum constant in direction. Thus, E is

correct.

Since B, C, D, and E are correct, the correct option is (4).

**4.** **(90)** Consider a small element of length $dx$ on the wire at a distance $x$ from the point charge $q$.

The linear charge density of the wire is $\lambda = \frac{Q}{L} = \frac{24\times10^{-6}}{0.1} = 240 \times 10^{-6}$ C/m.

The charge on the small element is $dQ = \lambda dx$.

The electrostatic force between the point charge $q$ and the element $dQ$ is $dF = \frac{kqdQ}{x^2} = \frac{kq\lambda dx}{x^2}$.

The total force $F$ is obtained by integrating $dF$ from $x = a$ to $x = a + L$, where $a = 2$ cm and $L = 10$ cm.

$F = \int_{0.02}^{0.12} \frac{kq\lambda}{x^2} dx = kq\lambda \left[-\frac{1}{x}\right]_{0.02}^{0.12}$

$F = kq\lambda \left(\frac{1}{0.02} - \frac{1}{0.12}\right) = kq\frac{Q}{L}\left(\frac{0.12-0.02}{0.02\times0.12}\right)$

$F = \frac{kqQ}{L}\frac{L}{a(a+L)} = \frac{kqQ}{a(a+L)}$

Substituting the given values: $k = 9 \times 10^9$ N.m$^2$/C$^2$, $q = 1 \times 10^{-6}$ C, $Q = 24 \times 10^{-6}$ C, $a = 0.02$ m, and $a + L = 0.12$ m.

$F = \frac{9\times10^9\times1\times10^{-6}\times24\times10^{-6}}{0.02\times0.12}$

$F = \frac{9\times24\times10^{-3}}{0.0024} = \frac{216\times10^{-3}}{2.4\times10^{-3}}$

$F = \frac{216}{2.4} = 90$ N.

**5.** **(2)** The electric potential at center O is the scalar sum of contributions from all 5 charges.

Each charge is equidistant from center, so $V = 5cdot\frac{kq}{r} = \frac{5q}{4\pi\epsilon_0 r}$.

The electric field is a vector sum of field vectors from all 5 charges.

Due to 5-fold rotational symmetry of a regular pentagon, the 5 field vectors point radially outward from center, equally spaced at 72° angles.

The vector sum of 5 equally spaced unit vectors is zero, so $\vec{E} = 0$.

**6.** **(3)** For three identical conducting bubbles with charge $q$ and radius $r$,

Initial potential on each: $V_i = \frac{kq}{r}$

When three bubbles coalesce,

Total charge $Q = 3q$ and new radius from volume: $\frac{4}{3}\pi R^3 = 3 \times \frac{4}{3}\pi r^3$

So $R = 3^{1/3}r$

Final potential: $V_f = \frac{k(3q)}{3^{1/3}r} = \frac{3kq}{3^{1/3}r} = V_i \cdot 3^{2/3}$

Ratio: $V_i : V_f = 1 : 3^{2/3}$

**7.** **(4)** Six point charges equally spaced 60° apart on a circle of radius R, with alternating arrangement creating a dipole-like configuration.

Each charge creates an electric field at the center of magnitude $E_0 = \frac{Q}{4\pi\varepsilon_0 R^2}$ pointing radially (outward for +Q, inward for -Q).

By vector addition of field contributions from all six charges with proper angle consideration:

The charges at specific positions create a net field with components that depend on their arrangement.

For the given symmetric configuration, the net electric field is: $\vec{E} = \frac{Q}{4\pi\varepsilon_0 R^2}(\sqrt{3}\hat{i} - \hat{j})$

This corresponds to magnitude $2E_0$ at angle of $-30^\circ$ from the positive x-axis.

**8.** **(1)** The charged bob experiences two forces at equilibrium:

Gravitational force: $F_g = mg$ (downward)

Electric force: $F_E = qE$ (horizontal, in direction of field)

These forces are perpendicular to each other. The string tension must balance their resultant.

At equilibrium, the string makes angle $\theta$ with vertical such that:

Vertical: $T\cos\theta = mg$

Horizontal: $T\sin\theta = qE$

The net force from gravity and electric field is $\sqrt{(mg)^2 + (qE)^2}$

Therefore, the tension in the string is: $T = \sqrt{m^2g^2 + q^2E^2}$

**9.** **(2)** The electric field is related to potential by $\vec{E} = -\nabla V$.

Given: $\vec{E} = Ax\hat{i} + By\hat{j}$ with $A = 10$ V/m² and $B = 5$ V/m²

This means:

$\frac{\partial V}{\partial x} = -Ax$ and $\frac{\partial V}{\partial y} = -By$

Integrating:

$V(x, y) = -\frac{Ax^2}{2} - \frac{By^2}{2} + C = -5x^2 - \frac{5y^2}{2} + C$

Using the boundary condition at $(10, 20)$ where $V = 500$ V:

$500 = -5(100) - \frac{5(400)}{2} + C$

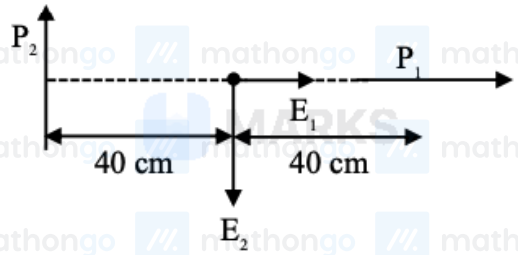
$500 = -500 - 1000 + C$

$C = 2000$ V

At the origin $(0, 0)$:

$V(0, 0) = -5(0) - \frac{5(0)}{2} + 2000 = 2000$ V

**10.**




**(1)**

$\vec{E}_2 = -\dfrac{KP_2}{r^3}$ ; $\vec{E}_1 = -\dfrac{2KP_1}{r^3}$

$P_1 = 2 \times 10^{-6} \times 10^{-2} = 2 \times 10^{-8}$

$P_2 = 4 \times 10^{-6} \times 10^{-2} = 4 \times 10^{-8}$

$\vec{E}_{net} = \dfrac{2 \times 9 \times 10^9 \times 2 \times 10^{-8}}{(0.4)^3}\hat{i} - \dfrac{9 \times 10^9 \times 4 \times 10^{-8}}{(0.4)^3}\hat{j}$

$\vec{E}_{net} = \dfrac{9 \times 10^9 \times 4 \times 10^{-8}}{(0.4)^3}[\hat{i} - \hat{j}]$

$|\vec{E}_{net}| = \dfrac{9 \times 10^4}{16}(\sqrt{2})$

**11.** **(1)** The total electrostatic energy $U$ of a system of two charges $q_1$ and $q_2$ in an external field is given by $U = q_1V(r_1) + q_2V(r_2) + \frac{kq_1q_2}{r_{12}}$.

Given external electric field $E = \frac{A}{r^2}\hat{r}$, the potential $V(r)$ is found by $V(r) = -\int_\infty^r E \cdot dr = -\int_\infty^r \frac{A}{r^2}dr = \frac{A}{r}$.

Parameters given:

$q_1 = 7\mu C = 7 \times 10^{-6}C$ at $r_1 = 9cm = 0.09m$.

$q_2 = -2\mu C = -2 \times 10^{-6}C$ at $r_2 = 9cm = 0.09m$.

Distance between charges $r_{12} = 9 - (-9) = 18cm = 0.18m$.

$A = 9 \times 10^5 N/C \cdot m^2$.

Calculating individual terms:

$q_1V(r_1) = q_1\frac{A}{r_1} = (7 \times 10^{-6})\frac{9\times10^5}{0.09} = 7 \times 10^{-6} \times 10^7 = 70J$.

$q_2V(r_2) = q_2\frac{A}{r_2} = (-2 \times 10^{-6})\frac{9\times10^5}{0.09} = -2 \times 10^{-6} \times 10^7 = -20J$.

Interaction energy $\frac{kq_1q_2}{r_{12}} = \frac{9\times10^9\times(7\times10^{-6})\times(-2\times10^{-6})}{0.18} = \frac{-126\times10^{-3}}{0.18} = -0.7J$.

Total energy $U = 70 - 20 - 0.7 = 49.3J$.

**12.** **(4)** Distance from origin to A: $r_A = \sqrt{4^2 + 4^2 + 2^2} = 6$ m

Distance from origin to B: $r_B = \sqrt{2^2 + 2^2 + 1^2} = 3$ m

Work done: $W = q_2(V_B - V_A) = q_2 \times \frac{q_1}{4\pi\varepsilon_0}\left(\frac{1}{r_B} - \frac{1}{r_A}\right)$

$= 2 \times 10^{-6} \times 9 \times 10^9 \times 10^{-8} \times \left(\frac{1}{3} - \frac{1}{6}\right)$

$= 18 \times 10^{-5} \times \frac{1}{6} = 30 \times 10^{-6}$ J

**13.** **(3)** Two point charges of 1 nC and 2 nC are placed at two corners of an equilateral triangle with side 3 cm. The work done in bringing a 3 nC charge from infinity to the third corner equals the charge times the potential at that point.

Both source charges are at distance 0.03 m from the third corner.

The potential due to the 1 nC charge is $V_1 = \frac{9\times10^9\times1\times10^{-9}}{0.03} = 300$ V

and due to the 2 nC charge is $V_2 = \frac{9\times10^9\times2\times10^{-9}}{0.03} = 600$ V.

Total potential: $V_{total} = 300 + 600 = 900$ V

Work done: $W = 3 \times 10^{-9} \times 900 = 2.7 \times 10^{-6}$ J = 2.7 μJ

**14.** **(3)** The potential at any point due to a charged spherical shell of radius $R$ and charge $Q$ is $V = \frac{1}{4\pi\epsilon_o}\frac{Q}{R}$ for points inside or on the surface ($r \le R$), and $V = \frac{1}{4\pi\epsilon_o}\frac{Q}{r}$ for points outside ($r > R$).

Let $k = \frac{1}{4\pi\epsilon_o}$. The potentials of the three shells $A, B, C$ with radii $a < b < c$ and charges $q_1, q_2, q_3$ are calculated by superposition.

For shell $A$ ($r = a$):

Shell $A$ is on its own surface, while it is inside shells $B$ and $C$.

$V_A = k\frac{q_1}{a} + k\frac{q_2}{b} + k\frac{q_3}{c} = \frac{1}{4\pi\epsilon_o}\left(\frac{q_1}{a} + \frac{q_2}{b} + \frac{q_3}{c}\right)$

For shell $B$ ($r = b$):

Shell $B$ is outside shell $A$, on its own surface, and inside shell $C$.

$V_B = k\frac{q_1}{b} + k\frac{q_2}{b} + k\frac{q_3}{c} = \frac{1}{4\pi\epsilon_o}\left(\frac{q_1+q_2}{b} + \frac{q_3}{c}\right)$

For shell $C$ ($r = c$):

Shell $C$ is outside shells $A$ and $B$, and on its own surface.

$V_C = k\frac{q_1}{c} + k\frac{q_2}{c} + k\frac{q_3}{c} = \frac{1}{4\pi\epsilon_o}\left(\frac{q_1+q_2+q_3}{c}\right)$

Comparing these expressions with the given options, option (3) matches correctly.

**15. (1)** The dipole moment is given by:

$\vec{p} = q_1\vec{r}_1 + q_2\vec{r}_2 + q_3\vec{r}_3$

$\vec{p} = (2q)(-3a)\hat{j} + (3q)(2a)\hat{i} + (-4q)(-2a)\hat{i}$

$\vec{p} = -6qa\hat{j} + 6qa\hat{i} + 8qa\hat{i}$

$\vec{p} = 14qa\hat{i} - 6qa\hat{j}$

$\vec{p} = 2qa(7\hat{i} - 3\hat{j})$

**16. (2)** From $V = ar^3 + b$, the electric field is $E = -dV/dr = -3ar^2$.

By Gauss's law:

$E(r) \cdot 4\pi r^2 = Q(r)/\epsilon_0$, so $-3ar^2 \times 4\pi r^2 = Q(r)/\epsilon_0$.

This gives $Q(r) = -12\pi a r^4 \epsilon_0$.

At $r = 1$: $Q(1) = -12\pi a\epsilon_0$.

Since $Q(1) = \alpha \times \pi a\epsilon_0$, we have $\alpha = -12$